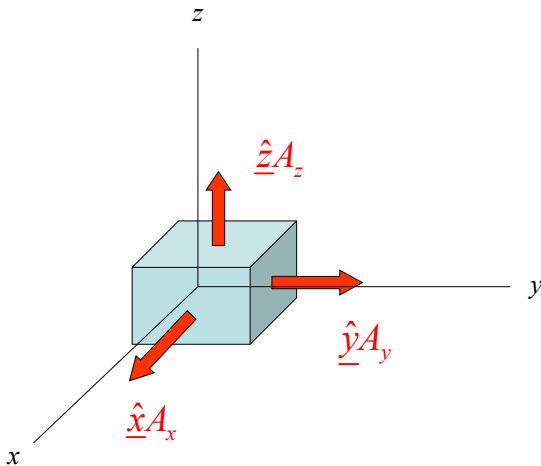


ECE 3317

Dr. Jiefu Chen

Notes 3

Review of Vector Calculus



Adapted from notes by Prof. Stuart A. Long

“Del” Operator

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

This is an “operator.”

Gradient

$$\nabla \phi = \hat{x} \frac{\partial \phi}{\partial x} + \hat{y} \frac{\partial \phi}{\partial y} + \hat{z} \frac{\partial \phi}{\partial z} \quad (\text{Vector})$$

Laplacian

$$\nabla^2 \phi = (\nabla \cdot \nabla) \phi = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi \quad (\text{Scalar})$$

“Del” Operator (cont.)

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

Vector $\underline{A}$: $\underline{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$

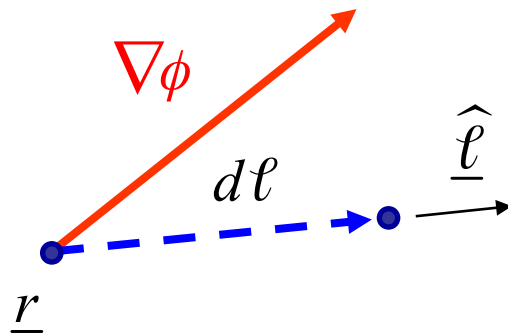
Divergence $\nabla \cdot \underline{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$ (Scalar)

Curl $\nabla \times \underline{A} = \hat{x} \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] + \hat{y} \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] + \hat{z} \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right]$
(Vector)

Gradient

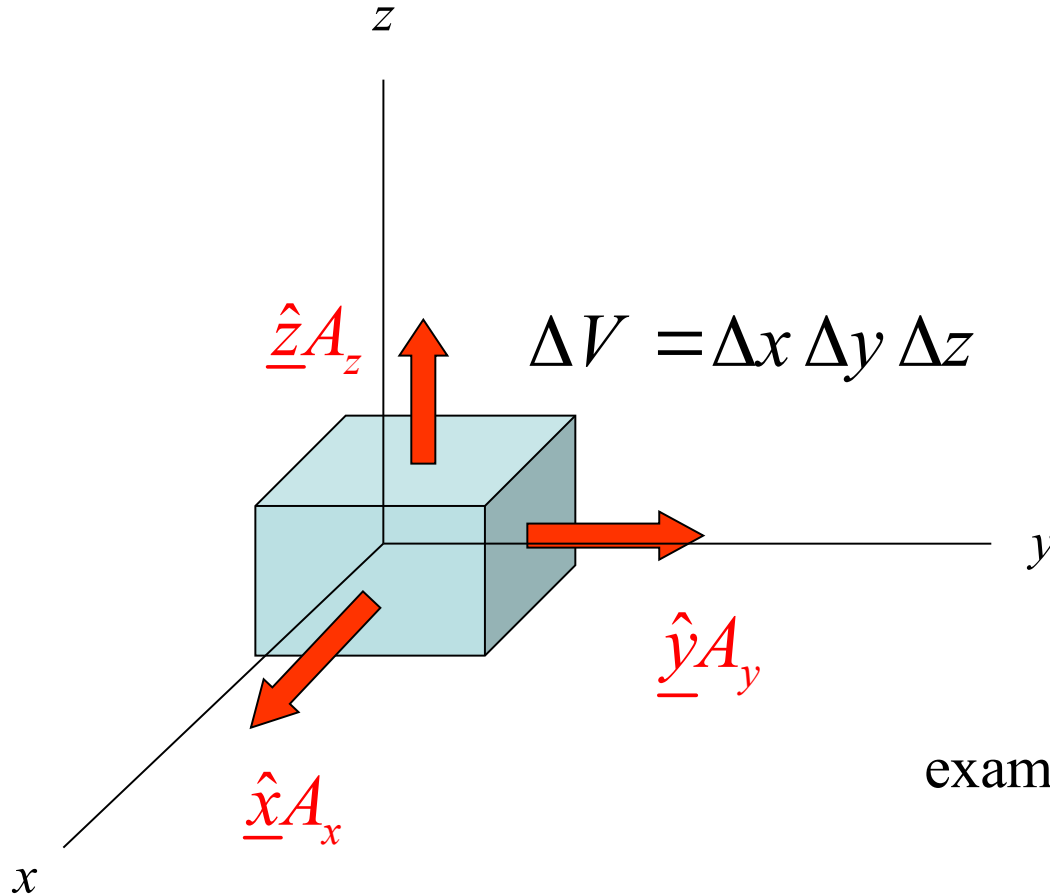
$$\nabla\phi = \hat{x}\left(\frac{\partial\phi}{\partial x}\right) + \hat{y}\left(\frac{\partial\phi}{\partial y}\right) + \hat{z}\left(\frac{\partial\phi}{\partial z}\right)$$

The gradient vector tells us the direction of maximum change in a function.



$$\frac{d\phi}{d\ell} = \nabla\phi \cdot (\hat{\underline{\ell}})$$

Divergence Operator



$$\nabla \cdot \underline{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \cdot \underline{A} \equiv \lim_{\Delta V \rightarrow 0} \frac{\oint \underline{A} \cdot \underline{\hat{n}} dS}{\Delta V}$$

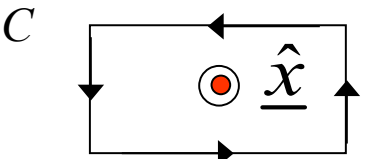
examples: $\underline{A} = \underline{\hat{x}}$; $\underline{A} = x\underline{\hat{x}}$; $\underline{A} = -x\underline{\hat{x}}$

The divergence measures the rate at which flux emanates from a region of space.

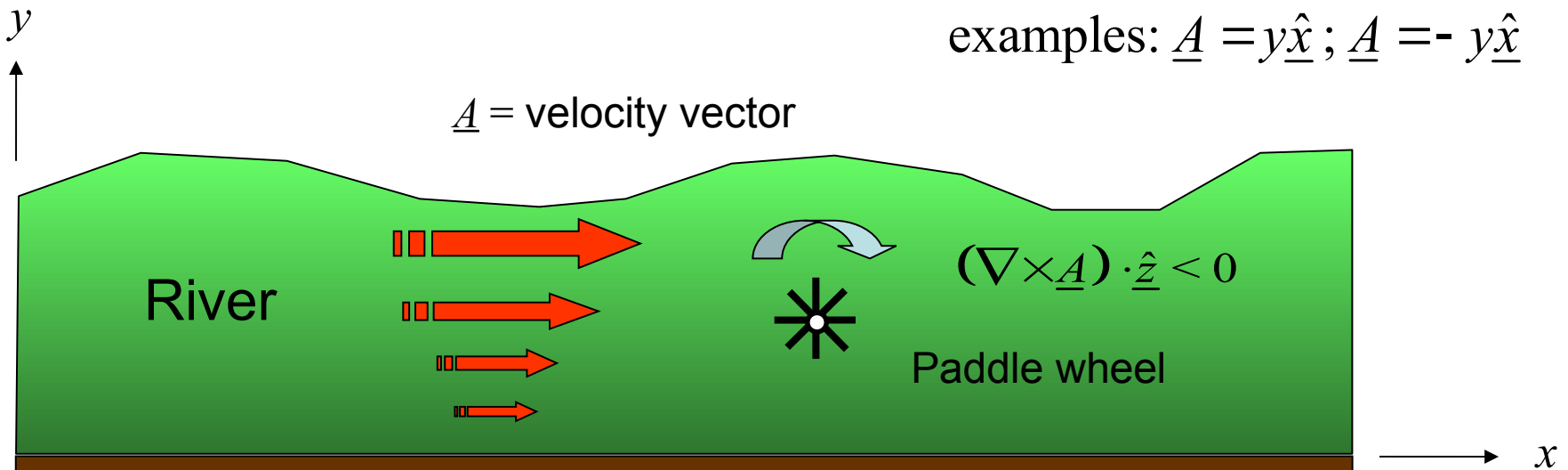
Curl Operator

$\Delta y, \Delta z \rightarrow 0$

$$\nabla \times \underline{A} = \hat{x} \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] + \hat{y} \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] + \hat{z} \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right]$$


 Δy "right-hand rule"

$$(\nabla \times \underline{A}) \cdot \hat{x} \equiv \lim_{\Delta y, \Delta z \rightarrow 0} \frac{\oint_C \underline{A} \cdot d\underline{r}}{\Delta y \Delta z}$$



A component of the curl tells us the rotation about that axis.

Vector Identities

Two fundamental “zero” identities:

$$\nabla \cdot (\nabla \times \underline{A}) = 0$$

$$\nabla \times (\nabla \phi) = \underline{0}$$

Vector Identities (cont.)

Another useful identity:

$$\nabla \cdot (\underline{A} \times \underline{B}) = \underline{B} \cdot (\nabla \times \underline{A}) - \underline{A} \cdot (\nabla \times \underline{B})$$

This will be useful in the derivation of the **Poynting theorem**.